

INT108: Python Programming & Problem Solving

100-Percentile Comprehensive Master Study Guide, 15 Practicals & Resource Dossier

Author / Lead: Saikat Koner (Black Hat Coders 108) | Target: 100/100 Marks (Grade 'O') | Course Code: INT108

1. Standard Reference Textbooks & Deep-Dive Chapter Context

Subject Area	Standard Reference Textbook	Core Chapter Concept & Key Exam Takeaways
Units I & II: Python Env & Flow	Kenneth A. Lambert (Cengage Learning)	Ch 1, 2 & 3: PVM compilation model (.py -> .pyc -> PVM), dynamic typing, variable names & keywords, integer vs float division // vs /, PEMDAS precedence (2**3**2=512), short-circuiting, while/for loops with else clauses, random module.
Unit III: Sequences & Data Structs	Reema Thareja (Oxford University Press)	Ch 5, 6 & 7: String slicing s[start:stop:step], s[::-1] reversal, list mutability & in-place methods, list comprehensions, shallow vs deep copy, tuple immutability, dict hashing O(1), sparse matrices representation.
Units IV & V: Functions & OOP	Reema Thareja / Kenneth A. Lambert	Ch 8, 10 & 11: Pass-by-object-reference, LEGB scoping rule (global vs nonlocal), *args, **kwargs, mutable default argument trap, recursion call stack, OOP 4 pillars, classes, __init__, self, Name Mangling (_Class__priv), C3 MRO & super(), dunder operator overloading, __slots__.
Unit VI: File I/O, Pickle, Regex	Kenneth A. Lambert (Cengage Learning)	Ch 9 & Regex: File modes (r, w, a, r+, wb), context manager with open(), object serialization via pickle.dump()/load(), security hazards of untrusted pickling, try-except-else-finally lifecycle, re module tokens & findall() pattern matching.

2. Curated YouTube Practical Video Hub & Video Demonstrations

Unit / Practical Area	Recommended Channel	Search Query / Video Reference	Core Practical Takeaway
Units 1-2: Python Setup & Flow	Corey Schafer	Corey Schafer Python Tutorial Beginners	PVM bytecode model, loops with else clauses.
Unit 3: Lists, Dicts & Slicing	Corey Schafer	Corey Schafer Python Lists Tuples Dicts	Slicing tricks, dict hashing O(1), copy traces.
Units 4-5: Functions & OOP	Telusko / Corey Schafer	Corey Schafer Python OOP Tutorials	LEGB scope, closures, decorators, C3 MRO, dunder.
Unit 6: File I/O, Pickle & Regex	Corey Schafer	Corey Schafer Python Regular Expressions	with open(), pickle.dump/load, regex findall.
All 15 University Practicals	Gate Smashers	Python Programming Lab Practicals Gate Smashers	Step-by-step code execution for Exp 1-15.
Master Viva Voce Preparation	Gate Smashers	Python Programming Viva Voce Gate Smashers	Top 75 examiner questions & code outputs.

3. High-Yield Theory Summary, 15 Practicals & Architecture Invariants

Units I & II (PVM, Types, Memory & Loops): Python source (.py) compiles to Bytecode (.pyc) executed on the PVM. In Python, variables are untyped symbolic references bound to typed heap objects. Exponentiation '**' has right-to-left associativity (2**3**2 = 512). The loop 'else' clause executes if and only if the loop completes cleanly without hitting a break. CPython uses Reference Counting + Cyclic GC.

Unit III (Sequences & Dictionaries): Strings and Tuples are immutable; Lists and Dictionaries are mutable. Slicing s[start:stop:step] with s[::-1] creates a reversed copy. Dictionary lookups operate in O(1) average time via compact Hash Tables. Sparse matrices are stored with coordinate tuple keys {(r, c): val}.

Units IV & V (Functions, Recursion & OOP): Python uses Pass-by-Object-Reference. LEGB rule dictates scope lookup. Default mutable arguments (def f(a=[])) persist across calls—fix with a=None. Classes use __init__ constructor, self reference, and Name Mangling (_Class__priv) for private attributes. C3 Linearization dictates MRO. Overload operators via dunder (__str__, __add__, __len__).

Unit VI (Files, Pickle & Regex): Always use 'with open()' as f: for guaranteed cleanup. Pickling (pickle.dump / pickle.load) handles binary serialization (untrusted pickling is an RCE hazard). Exception lifecycle: try -> except -> else (no error) -> finally (always). Regex meta-characters (\d, \w, \s, ^, \$, +, *, ?,) enable robust text parsing.

4. Top High-Frequency Viva Voce Scoring Blueprint

Q1: Is Python interpreted or compiled? — Ans: Both. Source (.py) is compiled to bytecode (.pyc), which is executed by the Python Virtual Machine (PVM).
Q2: What is the LEGB Rule? — Ans: The scope lookup hierarchy in Python: Local → Enclosing → Global → Built-in.
Q3: What is the difference between 'is' and '=='? — Ans: '==' checks value equality (content); 'is' checks memory identity (id(a) == id(b)).
Q4: What is the Mutable Default Argument Trap? — Ans: Default list [] is created once at function definition time and shared across calls. Fix with arg=None.
Q5: What is Pickling and Unpickling? — Ans: Pickling (pickle.dump) serializes Python objects to binary; Unpickling (pickle.load) restores objects from binary. Warning: RCE risk on untrusted data.
Q6: What is the Global Interpreter Lock (GIL)? — Ans: A mutex in CPython that restricts bytecode execution to one native thread at a time, ensuring thread-safe reference count management.